

EUNJI HWANG

Statistics & CS Graduate · Machine Learning, Predictive Modeling & Statistical Analysis

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SUMMARY

Statistics & Computer Science graduate (UBC) who builds and rigorously validates machine-learning and Bayesian models end-to-end — from feature engineering through inference, model comparison, and clear technical reporting in Python, R, and Stan, consistently proving which model wins and why.

TECHNICAL SKILLS

Machine Learning: Predictive Modeling, Classification, Regression, Bayesian & Probabilistic Modeling, Model Validation (LOO-CV, cross-validation), Feature Engineering, A/B Testing & Experiment Design

Statistics: Hypothesis Testing, Statistical Inference, Time-Series Forecasting, Data Visualization

Programming & Databases: Python (pandas, NumPy), R, Stan, SQL (MySQL), Java

ML Tools & Libraries: scikit-learn, Matplotlib, tidyverse, tidymodels, ggplot2, Jupyter, Git, Linux

Languages: English (fluent), Mandarin Chinese (fluent), Korean, Spanish

QUANTITATIVE PROJECTS

Bayesian Heavy-Tailed Modeling of Wildfire PM2.5 (BC) | *R, Stan* | [repo](#) Solo

- Single-handedly designed a Bayesian seasonal model of 4 years of daily PM2.5 (~1,460 observations) in Stan, showing a Student-*t* model captures the extreme wildfire-pollution tail risk a standard Normal model underestimates.
- Drove model selection with HMC/NUTS inference, full convergence diagnostics ($\hat{R}$, ESS, MCSE), and LOO-CV — the Student-*t* model won decisively ($\Delta\text{elpd} = 57.7$, ≈ 3.9 SE) and recovered heavy tails ($\hat{\nu} \approx 6.7$), confirmed by posterior predictive checks.
- Stress-tested every conclusion with a 50-replication synthetic-data calibration study and prior sensitivity analysis, and benchmarked HMC against variational inference (ADVI) to pinpoint where fast approximations break down.

Heart Disease Prediction | *R* | [repo](#) Classification

- Built and tuned a classification model in R (tidymodels) that detects heart disease at 90.2% sensitivity, deliberately optimizing recall to minimize missed diagnoses.
- Owned the full pipeline end-to-end — data cleaning, feature engineering, training, and evaluation — using tidyverse and ggplot2 to compare candidate models and surface the strongest performer.

Commercial Banking Outreach Analysis | *R* | [repo](#) Statistical Inference

- Applied bootstrapping and t-tests (difference in means/medians) to quantify how strongly call duration drove outreach success, backed by p-values and confidence intervals.
- Converted the results into a clear, evidence-based recommendation and presented the methodology and findings to an audience of 150.

Hot-Pot Restaurant Management System | *React, Node.js, MySQL* | [repo](#) CPSC 304

- Built (team of 3) a full-stack inventory-and-sales system across 6 restaurant branches on a normalized 20+ table MySQL schema designed from an ER diagram.
- Built the React frontend (axios) surfacing data-driven insights — least-profitable dish per branch, highest-profit day — backed by SQL joins, nested subqueries, and aggregation (GROUP BY / HAVING).

EXPERIENCE

Team Lead (Level 2) May 2023 – May 2026
Haidilao Hot Pot New Westminster, BC

- Promoted to Level-2 Team Lead; oversaw service for 50+ customer groups per shift while training and coordinating the team through peak hours.
- Diagnosed recurring bottlenecks and redesigned service workflows, improving turnaround and consistency.

EDUCATION

University of British Columbia Sept 2021 – June 2026
BSc, Statistics (Computer Science specialization) Vancouver, BC

- Relevant coursework:** Applied Machine Learning, Bayesian Statistics, Statistical Inference, Statistical Modelling, Databases, Data Science, Data Structures & Algorithms, Linear Algebra